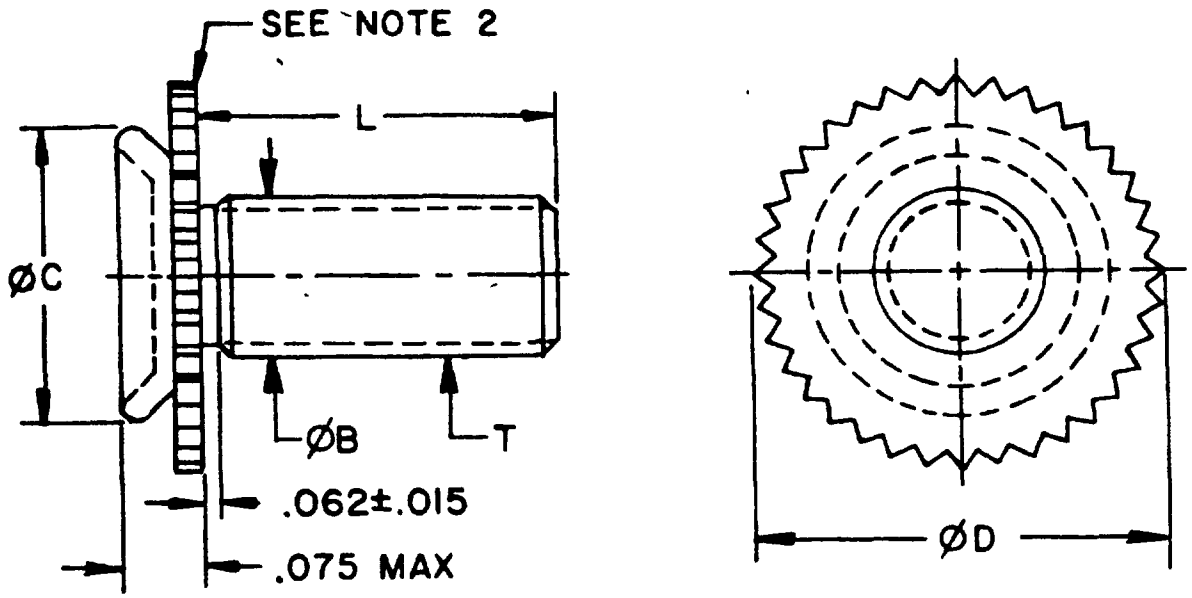
MILITARY SPECIFICATION SHEET

STUD, SELF-LOCKING, CONCEALED HEAD, KNURLED CLINCH TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the studs described herein shall consist of this specification and the latest issue of MIL-S-63540.



NOTES:

1. All dimensions are in inches.
2. Clinching profile may vary.
3. Blind holes may be deeper in panels thicker than the minimum with no adverse effect on the performance of stud.

FIGURE 1. STUD, SELF-LOCKING, CONCEALED HEAD, KNURLED CLINCH TYPE

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TABLE I. PART DIMENSIONS

T THREAD SIZE CLASS 2A	Ø B STUD DIA	Ø C MAX	Ø D ±.010
.112-40 UNC (4-40)	.112	.171	.205
.138-32 UNC (6-32)	.138	.212	.250
.164-32 UNC (8-32)	.164	.289	.328
.190-24 UNC (10-24)	.190	.311	.350
.190-32 UNF (10-32)	.190	.311	.350
.250-20 UNC (1/4-20)	.250	.343	.385

TABLE II. THREAD SIZE AND LENGTH DASH NUMBERS

L ±.015	T THREAD SIZE CLASS 2A					
	.112-40 UNC	.138-32 UNC	.164-32 UNC	.190-24 UNC	.190-32 UNF	.250-20 UNC
	THREAD SIZE AND LENGTH DASH NUMBER					
.250	1	6	12			
.375	2	7	13	18	23	
.500	3	8	14	19	24	
.625	4	9	15	20	25	28
.750	5	10	16	21	26	29
1.000		11	17	22	27	30

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REQUIREMENTS:

1. Material. Corrosion resistant steel shall be Type XM-7 (UNS S30430) in accordance with ASTM A 493 or Types 303 (UNS S30300) and 303Se (UNS S30323) in accordance with ASTM A 582.

2. Surface treatment. Corrosion resistant steel studs shall be cleaned, descaled and passivated in accordance with ANSI/ASTM A 380.

3. Hardness. Corrosion resistant steel studs shall have a hardness of 171HB minimum.

4. Pull-out and torque-out values. When the stud is properly installed, (see installation procedures), pull-out and torque-out values shall be as specified in Table III.

TABLE III. PULL-OUT AND TORQUE-OUT VALUES.

T THREAD SIZE CLASS 2A	SHEET THICK-NESS MIN (REF)	PULL-OUT LOAD (LBS) MIN 1/	TORQUE-OUT LOAD (IN LBS) MIN 1/	PULL-OUT LOAD (LBS) MIN 1/	TORQUE-OUT LOAD (IN LBS) MIN 1/
		ALUMINUM ALLOY SHEET MATERIAL 2/		COLD ROLLED STEEL SHEET MATERIAL 2/	
.112-40 UNC	.093	200	8	240	8
.138-32 UNC		260	13	350	13
.164-32 UNC		310	26	440	26
.190-24 UNC		360	48	680	48
.190-32 UNF		360	48	680	48
.250-20 UNC		400	115	700	115

1/ The load values shown are the minimum non-destruct values that apply during testing.

2/ Sheet material hardness values shall not exceed 70HRB.

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5. Part number. The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table II.

Example: M63540/4-12

Dash number

Specification sheet number

M63540/4-12 indicates - Stud, self-locking, concealed head, knurled clinch type; CRES; .164-32 UNC-2A thread; .250 length; .289 head diameter.

INSTALLATION PROCEDURES:

a. Mill or drill blind hole in sheet material with diameter as shown in Table IV to a minimum depth of .075.

b. Locate head of stud squarely in milled or drilled hole in the mounting surface with anvil below sheet as shown (see Figure 2).

c. Apply force to embed the knurled ring flush with sheet material (see Figure 2).

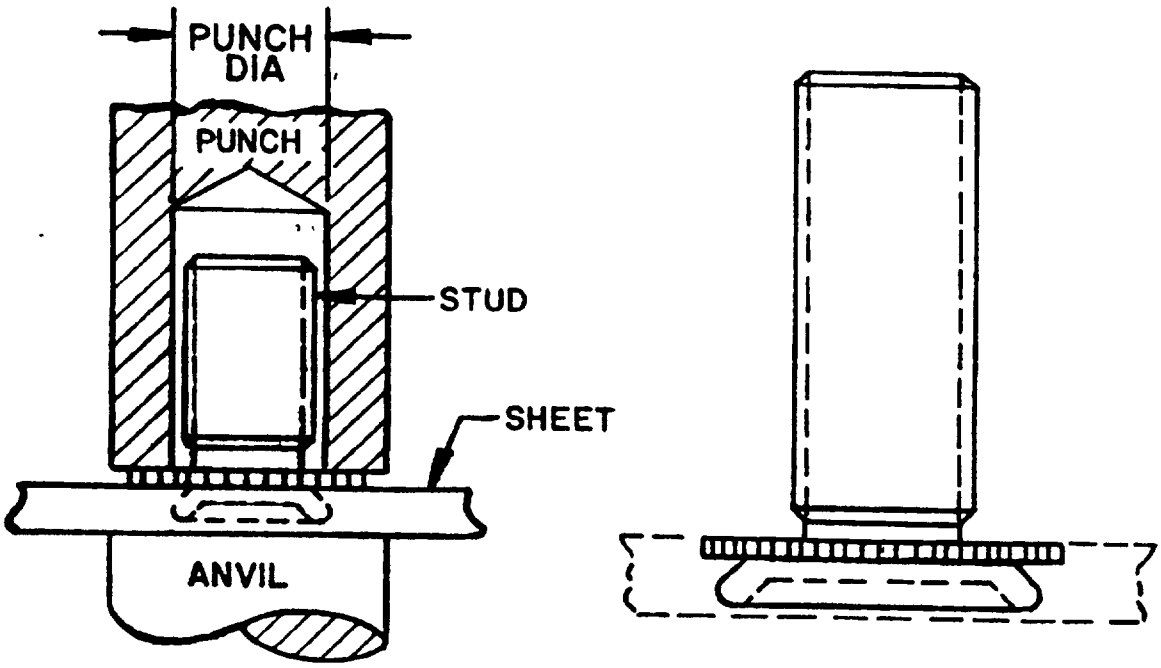


FIGURE 2. GENERAL TOOLING SETUP.

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TABLE IV. INSTALLATION DIMENSIONS

T THREAD SIZE CLASS 2A	PUNCH DIA +.003 -.000 (REF) 1/	SHEET THICKNESS MIN (REF)	BLIND MOUNTING HOLE +.003 -.000 (REF)	CENTER OF HOLE TO EDGE OF SHEET MIN (REF)	NUT TIGHTENING TORQUE (IN LBS) MAX (REF)
.112-40 UNC	.120	.093	.172	.156	4.75
.138-32 UNC	.150		.213	.188	9
.164-32 UNC	.180		.290	.219	18
.190-24 UNC	.203		.312	.250	24
.190-32 UNF	.203		.312	.250	32
.250-20 UNC	.261		.344	.281	80

1/ See Figure 2

Custodians:
Army - AR
Navy - MC
Air Force - 99

Review activities:
Army - AT, ME
Air Force - 11, 82
DLA - IS

User activities:
Army - ER
Navy - SH, OS
NSA - NS

Preparing activity:
Army - AR

Agent:
DLA - IS

(Project 5307-0334)

* GPO 1986- 605-036/42438

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